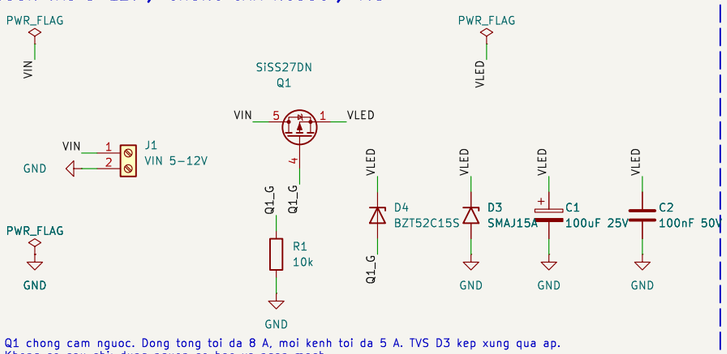
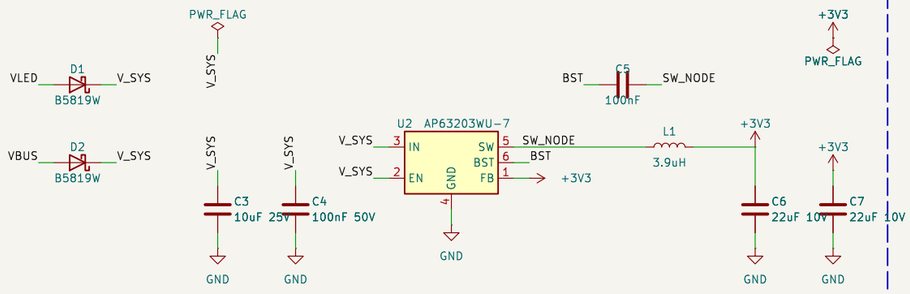
NGUON VAO 5–12V / CHONG CAM NGUOC / TVS

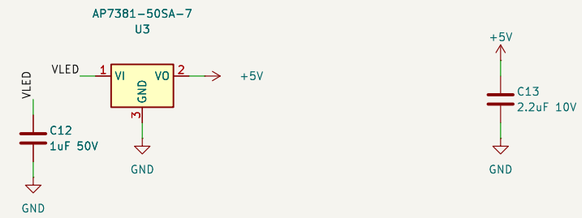


Q1 chong cam nguoc. Dong tong toi da 8 A, moi kenh toi da 5 A. TVS D3 kep xung qua ap.
Khong co cau chi: dung nguon co bao ve ngan mach.

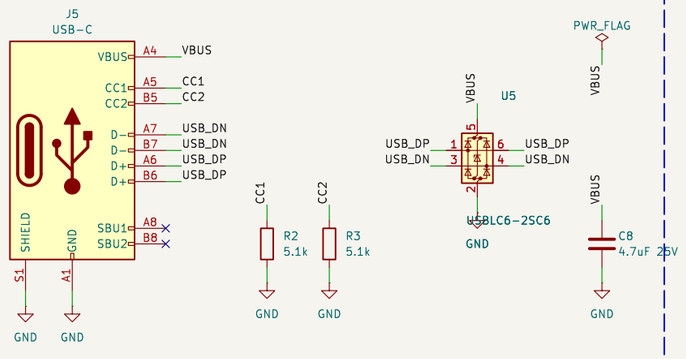
NGUON 3.3V (buck AP63203)



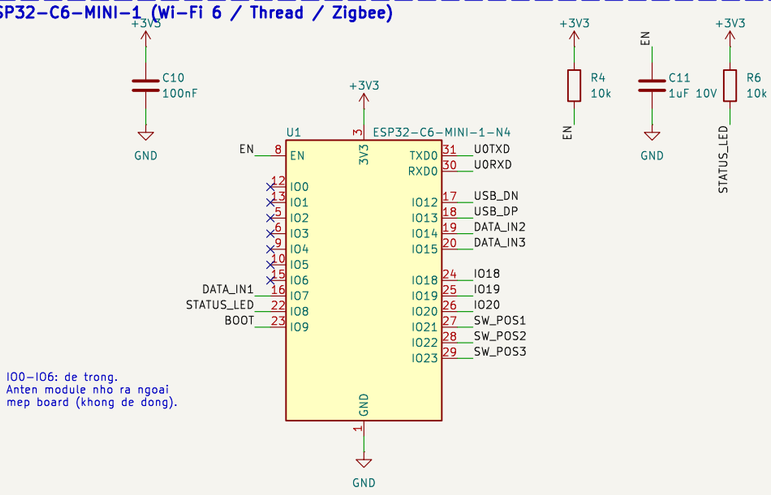
5V CHO IC DICH MUC (chi co khi co nguon LED)



USB-C: NAP CODE / DEBUG (USB Serial/JTAG)

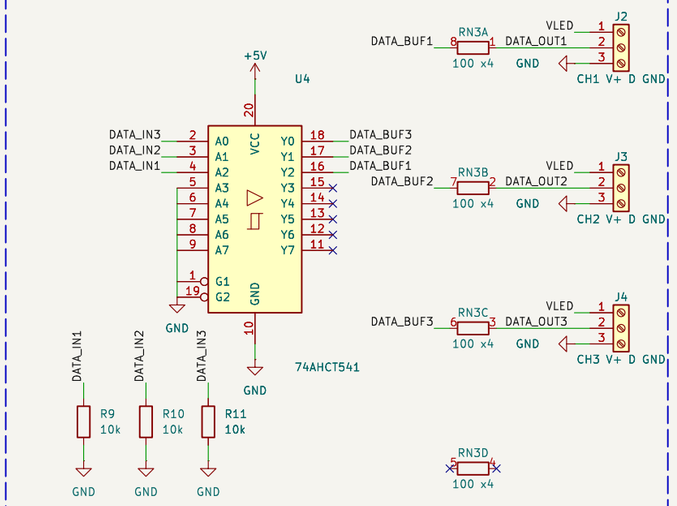


ESP32-C6-MINI-1 (Wi-Fi 6 / Thread / Zigbee)



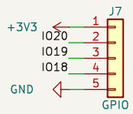
IO0–IO6: de trong.
Anten module nho ra ngoai
mep board (khong de dong).

DICH MUC 3.3V -> 5V + 3 KENH WS2812



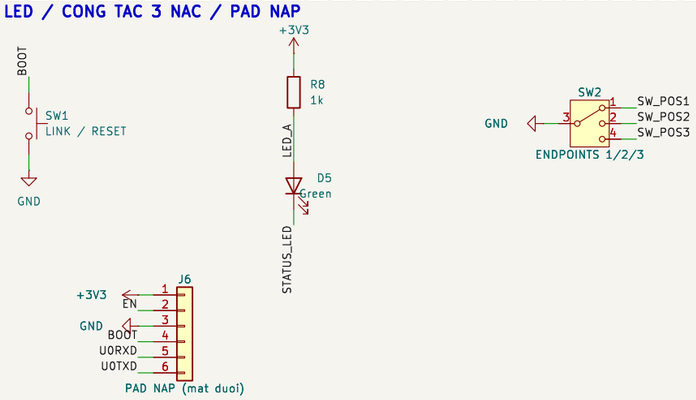
DATA 5V: WS2812B (5V) va WS2811 (12V) deu nhan muc 5V.

MO RONG: 3 GPIO TU DO (J7)



IO18/IO19/IO20: GPIO thuong, co pull-up yeu chi trong luc reset.
IO8 (LED, R6 keo len) va IO9 (nut, pull-up noi) la chan strap, khong dung cho J7.

NUT / LED / CONG TAC 3 NAC / PAD NAP



SW1 (GPIO9 = BOOT): bam nhanh = bat/tat hoac vao che do ket noi; giu >5 s = factory reset.
Giu SW1 trong luc cap nguon = vao che do nap ROM.

SW2 = C&K PCM13SMTR, chan 3 la chan chung (noi GND). Nac 1 noi 1–3, nac 2 noi 2–3, nac 3 noi 3–4.
Firmware: IO21/IO22/IO23 la input, bat pull-up noi; chan nao doc 0 -> so endpoint Matter = 1/2/3.
Khong chan nao doc 0 (dang gat giua 2 nac) -> giu gia tri cu.

Generated by tools/build_sch.py from tools/design.py
Input 5–12V, 3x WS2812/WS2811 outputs, 1 button, 3-position endpoint switch, 3 spare GPIOs

Haxeko

Sheet: /
File: c6_ws2812_3ch.kicad_sch

Title: ESP32-C6 WS2812 3-channel Matter LED controller

Size: A3	Date: 2026-11-09	**Rev: 2.0**
KiCad E.D.A. 9.0.9		Id: 1/1